

Work problem #1 in the space provided. Make sure to communicate mathematically how you are arriving at your answer even if you do a lot of the work in your head or on your calculator!! Find the answer to problem #1. Number the problem in the same box #2. Work the second problem. Continue working until all are completed.

Answer: $(x-5)^2$ #1 Simplify $\frac{(5x^3)^2 y^3}{15y^2}$.

$$\frac{5x^6 y^3}{15y^2} = \frac{5x^6 y}{3}$$

Answer: $y = -1.5x + 17$ # ____ Write the equation of the line if $m = -\frac{3}{2}$ and $b = 16$.

$$y = mx + b$$

$$y = -\frac{3}{2}x + 16$$

Answer: $5x - 3$ # ____ Factor (difference of squares) the binomial $25x^2 - 4$ and find one of its factors.

~~$$(5x-3)(5x+3)$$~~

$$(5x+2)(5x-2)$$

a

Answer: $\frac{15}{8}$ # ____ Solve for x: $\frac{x-1}{3} + \frac{x+5}{4} = \frac{1}{2}$

~~$$3x-1 + x+5 = 3 \times 0.5$$~~

~~$$4x-1x+5 = 1.5$$~~

$$\frac{-5}{7}$$

Answer: $\emptyset$ # 12 Solve for x: $\frac{3-4x}{6} - \frac{2x-3}{8} \geq 2-x$

$$\frac{12-16x}{24} - \frac{6x-9}{24} \geq 28-24x$$

$$[13.5, \infty)$$

Answer: -6# ____ Find c so that $x^2 - 10x + c$ is a perfect square trinomial. (Find c to complete the square.) Then, factor the trinomial.

~~$$x^2 - 10x + c$$~~

$$25$$

$$x^2 \cdot x^2 = x^4$$

$$y - y_1 = m(x - x_1)$$

Answer: $x=a$

Write the equation of the line perpendicular to $2x - 3y = 12$ which through the point $(16, -7)$.

$$2x - 3y = 12$$

$$y = \frac{2}{3}x - 4$$

$$-\frac{3}{2}$$

$$\frac{-3y}{-3} = \frac{-2x + 12}{-3}$$

$$-1 = -\frac{3}{2}$$

$$y = -2x - 4$$

$$-7 = -2(16) + b$$

$$-7 = -32 + b$$

$$b = 25$$

$$y = -2x + 25$$

Answer: $\frac{5x^6y}{3}$

2 Simplify $\frac{(x^{-2}y^{-1})^{-3}}{(y^6x^{-1})^{-2}}$

$$\frac{(x^{-2}y^{-1})^{-3}}{(y^6x^{-1})^{-2}} = \frac{x^6y^3}{y^{12}x^2} = \frac{x^4y^3}{y^{12}}$$

$$= \frac{x^4}{y^9}$$

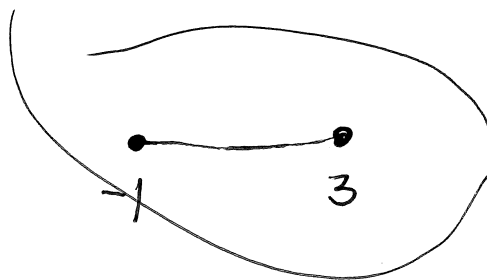
Answer: $(-\frac{1}{3}, -\frac{3}{4})$

18 Find the midpoint between -3π and π .

$$\frac{-3\pi + \pi}{2} = -\pi$$

Answer: $(-\infty, 6)$

Graph $-1 \leq x \leq 3$ on a number line. Put the answer in interval notation.



Answer: $5x+2$

3 Factor (bottoms up!) the trinomial $10x^2 - 13x - 3$ and find one of its factors.

$$(5x - 1)(2x + 3)$$

$$10x^2 - 13x - 3$$

Answer: $[13.5, \infty)$

Write the equation of the line which passes through the points $(4, 2)$ and $(-3, 1)$.

$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{1 - 2}{-3 - 4} = \frac{-1}{-7} = \frac{1}{7}$$

$$y = \frac{1}{7}x + b$$

$$2 = 4\left(\frac{1}{7}\right) + b$$

$$2 = \frac{4}{7} + b$$

$$b = 2 - \frac{4}{7} = \frac{10}{7}$$

$$y = \frac{1}{7}x + \frac{10}{7}$$

$$2 = 4\frac{1}{7} + b$$

Answer: x^4y^{15}

____ Find the distance between -5 and 17.

22

Answer: $2x - 6$

____ Solve for x by first setting up two equations. Then, find the PRODUCT of your solutions.

$$2x + 5 = 7$$

$$2x + 5 = 7$$

$$2x = 2$$

!

$$2x + 5 = -7$$

$$2x = -12$$

$$-1$$

(-7)

Answer: $-\frac{5}{7}$

____ Solve for x: $-2(x - 4) - \frac{1}{2}(3 - 4x) = -12$

$$-2x + 8 - 1.5 - 2x = -12$$

(0)

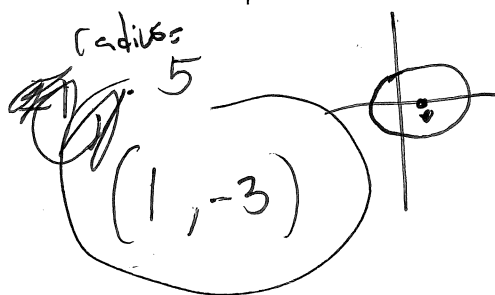
Answer: $y = -1.5x + 16$

2 Factor (GCF) the binomial $25x^2 - 15x$ and find one of its factors.

($5x - 3$)

Answer: $[-1, 3]$

____ The equation $(x - h)^2 + (y - k)^2 = r^2$ defines a circle with radius r and center (h, k). Find the center and radius AND graph the circle on the x-y coordinate plane. $(x - 1)^2 + (y + 3)^2 = 25$. Find the center to move on to the next question.



Answer: $-\pi$

____ If AB is a line segment with midpoint (2, 6) and endpoint A(5, 9), find the other endpoint, B.

A ——— B
5, 9 2, 6

B(3, 3)

(-1, 3)

Answer: $(-1, 3)$

~~16~~ Write $6 > x$ in interval notation.

~~$(-\infty, 6)$~~
 $(-\infty, 6)$

Answer: $7y - x = 10$

_____ Write the equation of the line which passes through the points $(a, 2b)$ and $(a, 3b)$.

$x = a$

Answer: $2\sqrt{13}$

9 Find the midpoint between $(-\frac{7}{3}, \frac{3}{4})$ and

$(\frac{5}{3}, -\frac{9}{4})$.

$(-\frac{1}{3}, -\frac{3}{4})$
 $-\frac{7}{3}, \frac{3}{4}$
 $\frac{5}{3}, -\frac{9}{4}$
 1.5

Answer: $5x + 1$

_____ What is the SUM of the factors of the trinomial $x^2 - 6x + 5$?

$(x + 1)(x - 5)$
 $(x - 1)(x - 5)$
 $x^2 - 5x - 1x$
 -6

Answer: $(1, -3)$

_____ Solve for x : $\frac{2}{3}x - \frac{1}{4} = 1$

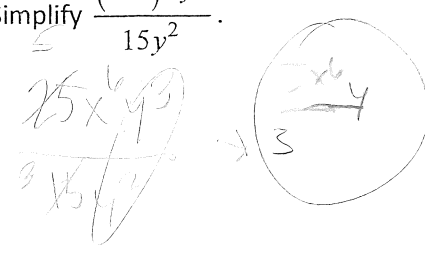
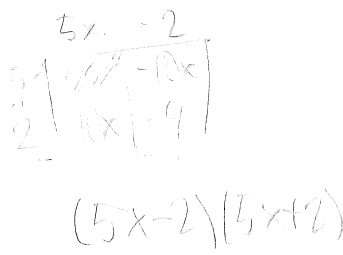
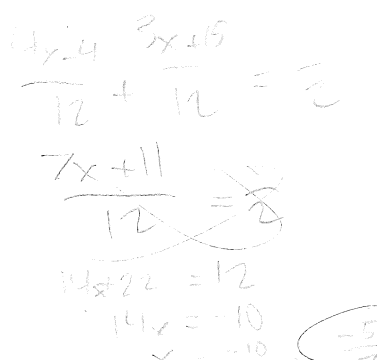
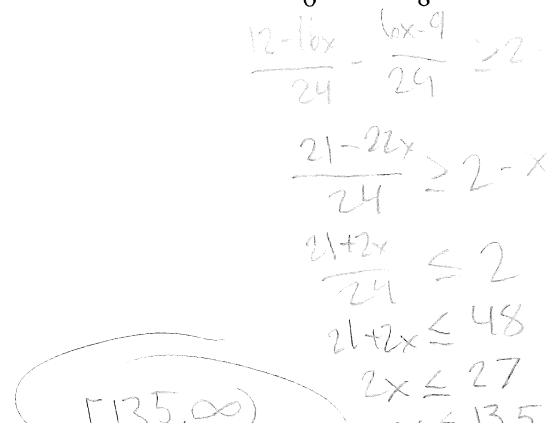
$\frac{2}{3}x = 1\frac{1}{4}$
 $1\frac{5}{4}$
 1.875

Answer: 22

13 Find the length of the line segment AB if the coordinates of the endpoints are $A(-1, 5)$ and $B(3, 11)$.

$2\sqrt{13}$
 $36 + 16$
 50
 $\sqrt{50}$
 $2\sqrt{13}$

Work problem #1 in the space provided. Make sure to communicate mathematically how you are arriving at your answer even if you do a lot of the work in your head or on your calculator!! Find the answer to problem #1. Number the problem in the same box #2. Work the second problem. Continue working until all are completed.

Answer: $(x-5)^2$ #1 Simplify $\frac{(5x^3)^2 y^3}{15y^2}$. 	Answer: $y = -1.5x + 17$ # <u>18</u> Write the equation of the line if $m = -\frac{3}{2}$ and $b = 16$. $y = -\frac{3}{2}x + 16$
Answer: $5x - 3$ # <u>20</u> Factor (difference of squares) the binomial $25x^2 - 4$ and find one of its factors. 	Answer: $\frac{15}{8}$ # <u>12</u> Solve for x: $\frac{x-1}{3} + \frac{x+5}{4} = \frac{1}{2}$ 
Answer: $\emptyset$ # <u>14</u> Solve for x: $\frac{3-4x}{6} - \frac{2x-3}{8} \geq 2-x$ 	Answer: -6 # <u>24</u> Find c so that $x^2 - 10x + c$ is a perfect square trinomial. (Find c to complete the square.) Then, factor the trinomial. $x^2 - 10x + 25$ $(x-5)^2$

Answer: $x=a$ # <u>17</u> Write the equation of the line perpendicular to $2x - 3y = 12$ which through the point $(16, -7)$. $2x - 3y = 12$ $3y = 2x - 12$ $y = \frac{2}{3}x - 4$ $y = -\frac{3}{2}x + b$ $-7 = -\frac{3}{2}(16) + b$ $-7 = -24 + b$ $17 = b$ $y = -\frac{3}{2}x + 17$	Answer: $\frac{5x^6y}{3}$ # <u>2</u> Simplify $\frac{(x^{-2}y^{-1})^{-3}}{(y^6x^{-1})^{-2}}$ $\frac{y^3}{y^{-12}x^2} \rightarrow x^4y^{15}$
Answer: $(-\frac{1}{3}, -\frac{3}{4})$ # <u>6</u> Find the midpoint between -3π and π. $-3\pi + \pi = -2\pi/2$ $-\pi$	Answer: $(-\infty, 6)$ # <u>9</u> Graph $-1 \leq x \leq 3$ on a number line. Put the answer in interval notation.  $[-1, 3]$
Answer: $5x+2$ # <u>21</u> Factor (bottoms up!) the trinomial $10x^2 - 13x - 3$ and find one of its factors. $5x \begin{array}{ c c } \hline 2x & -3 \\ \hline 10x^2 & -15x \\ \hline 1 & 2x \\ \hline \end{array} -3$ $(5x+1)(2x-3)$ $\begin{array}{r} 30x^2 \\ -15x \times 2x \\ \hline -13x \end{array}$	Answer: $[13.5, \infty)$ # <u>15</u> Write the equation of the line which passes through the points $(4, 2)$ and $(-3, 1)$. $\frac{1-2}{-3-4} = \frac{-1}{-7} \rightarrow \frac{1}{7}$ $y = \frac{1}{7}x + b$ $1 = \frac{1}{7}(-3) + b$ $1 = -\frac{3}{7} + b$ $1.43 = b$ $y = \frac{1}{7}x + 1.43$ $7y = x + 10$ $7y - x = 10$

Answer: x^4y^{15} # <u>3</u> Find the distance between -5 and 17. $\begin{array}{r} 17 \\ +5 \\ \hline 22 \end{array}$	Answer: $2x - 6$ # <u>23</u> Solve for x by first setting up two equations. Then, find the PRODUCT of your solutions. $2x + 5 = 7$ $\begin{array}{ll} 2x + 5 = 7 & 2x + 5 = -7 \\ 2x = 2 & 2x = -12 \\ x = 1 & x = -6 \\ & 1(-6) = -6 \end{array}$
Answer: $-\frac{5}{7}$ # <u>13</u> Solve for x: $-2(x - 4) - \frac{1}{2}(3 - 4x) = -12$ $-2x + 8 - \frac{3}{2} + 2x = -12$ No solution	Answer: $y = -1.5x + 16$ # <u>19</u> Factor (GCF) the binomial $25x^2 - 15x$ and find one of its factors. $(5x)(5x - 3)$
Answer: $[-1, 3]$ # <u>10</u> The equation $(x - h)^2 + (y - k)^2 = r^2$ defines a circle with radius r and center (h, k). Find the center and radius AND graph the circle on the x-y coordinate plane. $(x - 1)^2 + (y + 3)^2 = 25$. Find the center to move on to the next question. $h = 1 \quad k = -3$ Center: $(1, -3)$	Answer: $-\pi$ # <u>7</u> If AB is a line segment with midpoint (2, 6) and endpoint A(5, 9), find the other endpoint, B. $\begin{array}{l} 2(2, 6) \rightarrow (4, 12) \\ (4, 12) \\ - (5, 9) \\ \hline (-1, 3) \end{array}$

Answer: (-1, 3) # <u>8</u> Write $6 > x$ in interval notation. $(-\infty, 6)$	Answer: $7y - x = 10$ # <u>16</u> Write the equation of the line which passes through the points (a, 2b) and (a, 3b). $x = a$
Answer: $2\sqrt{13}$ # <u>5</u> Find the midpoint between $(-\frac{7}{3}, \frac{3}{4})$ and $(\frac{5}{3}, -\frac{9}{4})$. $(\frac{3}{4} + \frac{9}{4}) = \frac{12}{4} = 3$ $(-\frac{7}{3} + \frac{5}{3}) = -\frac{2}{3}$ $(-\frac{2}{3}, 3)$	Answer: $5x+1$ # <u>22</u> What is the SUM of the factors of the trinomial $x^2 - 6x + 5$? $(x-1)(x-5)$ $2x = 6$
Answer: (1, -3) # <u>11</u> Solve for x: $\frac{2}{3}x - \frac{1}{4} = 1$ $\frac{2}{3}x - \frac{1}{4} = 1$ $+\frac{1}{4} \quad +\frac{1}{4}$ $\frac{2}{3}x = 1 + \frac{1}{4}$ $\frac{2}{3}x = \frac{5}{4}$ $x = \frac{15}{8}$	Answer: 22 # <u>4</u> Find the length of the line segment AB if the coordinates of the endpoints are A(-1, 5) and B(3, 11). $6^2 + 4^2 = x^2$ $36 + 16 = x^2$ $52 = x^2$ $\sqrt{52} = x$ $2\sqrt{13} = x$

Work problem #1 in the space provided. Make sure to communicate mathematically how you are arriving at your answer even if you do a lot of the work in your head or on your calculator!! Find the answer to problem #1. Number the problem in the same box #2. Work the second problem. Continue working until all are completed.

Answer: $(x-5)^2$ #1 Simplify $\frac{(5x^3)^2 y^3}{15y^2}$. $\frac{(5x^3)^2 y^3}{15y^2} = \frac{25x^6 y^3}{15y^2} = \frac{5x^6 y}{3}$	Answer: $y = -1.5x + 17$ # <u>18</u> Write the equation of the line if $m = -\frac{3}{2}$ and $b = 16$. $y = -\frac{3}{2}x + 16$
Answer: $5x - 3$ # <u>20</u> Factor (difference of squares) the binomial $25x^2 - 4$ and find one of its factors. $\begin{array}{cc} 5x & 2 \\ \hline 5x & 25x^2 & 10x & 1 \\ -2 & -10x & -4 & \end{array}$	Answer: $\frac{15}{8}$ # <u>12</u> Solve for x: $\frac{x-1}{3} + \frac{x+5}{4} = \frac{1}{2}$ $\left(\frac{4}{4}\right) \frac{x-1}{3} + \frac{x+5}{4} \left(\frac{3}{3}\right) = \frac{1}{2}$ $\frac{4x-4}{12} + \frac{3x+15}{12} = \frac{1}{2}$ $\frac{7x+11}{12} = \frac{1}{2}$ $14x + 22 = 12$
Answer: $\emptyset$ # <u>14</u> Solve for x: $\frac{3-4x}{6} - \frac{2x-3}{8} \geq 2-x$ $\frac{2x+21}{24} - \frac{12-16x}{24} - \frac{6x-9}{24} \geq 2-x$ $2x \leq 27$ $x \leq 13.5$ $\frac{-22x+21}{24} \geq 2-x$ $\frac{2x+21}{24} \leq 2$ <u>$[13.5, \infty)$</u>	Answer: -6 # <u>24</u> Find c so that $x^2 - 10x + c$ is a perfect square trinomial. (Find c to complete the square.) Then, factor the trinomial. $\begin{array}{cc} x & -5 \\ \hline x & x^2 & -5x & \\ -5 & -5x & 25 & \end{array}$ $(x-5)^2$



Answer: $x=a$ # <u>17</u> Write the equation of the line perpendicular to $2x - 3y = 12$ which through the point $(16, -7)$. $3y = 2x - 12$ $y = \frac{2}{3}x - 4$ $y = -1.5x + 17$	Answer: $\frac{5x^6y}{3}$ # <u>2</u> Simplify $\frac{(x^{-2}y^{-1})^{-3}}{(y^6x^{-1})^{-2}}$ $x^6 y^3$ $y^{-12} x^2$ $x^4 y^{15}$
Answer: $(-\frac{1}{3}, -\frac{3}{4})$ # <u>6</u> Find the midpoint between -3π and π. $-3\pi + \pi = -2\pi$ $-2\pi / 2$ $-\pi$	Answer: $(-\infty, 6)$ # <u>9</u> Graph $-1 \leq x \leq 3$ on a number line. Put the answer in interval notation.  $[-1, 3]$
Answer: $5x+2$ # <u>21</u> Factor (bottoms up!) the trinomial $10x^2 - 13x - 3$ and find one of its factors. $\begin{array}{c cc} & 5x & 2 \\ \hline 2x & 10x^2 & 2x \\ -3 & -15x & -3 \end{array}$	Answer: $[13.5, \infty)$ # <u>15</u> Write the equation of the line which passes through the points $(4, 2)$ and $(-3, 1)$. $\frac{1-2}{-3-4} = \frac{-1}{-7} = \frac{1}{7}$ $2 = \frac{1}{7}(4) + b$ $b = 1.43$

Answer: x^4y^{15} # <u>3</u> Find the distance between -5 and 17. 22	Answer: $2x - 6$ # <u>23</u> Solve for x by first setting up two equations. Then, find the PRODUCT of your solutions. $2x + 5 = 7$ 10 - 6
Answer: $-\frac{5}{7}$ # <u>13</u> Solve for x: $-2(x - 4) - \frac{1}{2}(3 - 4x) = -12$ $-2x + 8 - 1.5 + 2x = -12$ $6.5 + 0x = -12$ $-6.5 \quad -6.5$ $0x = -18.5$ No solution	Answer: $y = -1.5x + 16$ # <u>19</u> Factor (GCF) the binomial $25x^2 - 15x$ and find one of its factors. $\begin{array}{r l} 5x & 25x^2 \\ \hline -3 & \end{array}$ $(5x)(5x - 3)$
Answer: $[-1, 3]$ # <u>10</u> The equation $(x - h)^2 + (y - k)^2 = r^2$ defines a circle with radius r and center (h, k). Find the center and radius AND graph the circle on the x-y coordinate plane. $(x - 1)^2 + (y + 3)^2 = 25$. Find the center to move on to the next question. r = 25. center = (1, -3)	Answer: $-\pi$ # <u>7</u> If AB is a line segment with midpoint (2, 6) and endpoint A(5, 9), find the other endpoint, B. $2(2, 6)$ $(4, 12)$ $-(5, 9)$ $(-1, 3)$

$$\frac{7y = x + 10}{7}$$

$$y = \frac{1}{7}x + 1.42$$

Answer: (-1, 3)

8 Write $6 > x$ in interval notation.

$$(-\infty, 6)$$

Answer: $7y - x = 10$ # 16 Write the equation of the line which passes through the points (a, 2b) and (a, 3b).

$$x = a$$

Answer: $2\sqrt{13}$ # 5 Find the midpoint between $(-\frac{7}{3}, \frac{3}{4})$ and

$$(\frac{5}{3}, -\frac{9}{4})$$

$$\frac{3}{4} + \frac{-9}{4} = \frac{-6}{4}$$

$$\frac{-7}{3} + \frac{5}{3} = \frac{-2}{3}$$

$$(-\frac{1}{3}, -\frac{3}{4})$$

Answer: $5x+1$ # 22 What is the SUM of the factors of the trinomial $x^2 - 6x + 5$?

$$\begin{array}{r|rr} x & -5 & \\ \hline x^2 & -5x & \\ -1 & -x & 5 \end{array}$$

Answer: (1, -3)

11 Solve for x: $\frac{2}{3}x - \frac{1}{4} = 1$

$$x = \frac{15}{8}$$

$$\frac{2}{3}x - \frac{1}{4} = 1$$

$$x = \frac{5}{4} \div \frac{2}{3}$$

$$\frac{5}{4} \times \frac{3}{2} = \frac{15}{8}$$

Answer: 22

4 Find the length of the line segment AB if the coordinates of the endpoints are A(-1, 5) and B(3, 11).

$$4^2 + 6^2 = x^2$$

$$16 + 36 = x^2$$

$$\sqrt{52} = \sqrt{x^2}$$

$$x = 2\sqrt{13}$$

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Answer: $5x - 3$ # <u>20</u> Factor (difference of squares) the binomial $25x^2 - 4$ and find one of its factors. <table border="1" style="margin-left: auto; margin-right: auto;"><tr><td>$5x$</td><td>$+2$</td></tr><tr><td>$-5x$</td><td>-2</td></tr></table>	$5x$	$+2$	$-5x$	-2	Answer: $\frac{15}{8}$ # <u>12</u> Solve for x: $\left(\frac{x-1}{3}\right)\left(\frac{x+5}{4}\right) = \left(\frac{1}{2}\right)$ $4(x-1) + 3(x+5) = 6$ $4x - 4 + 3x + 15 = 6$ $7x + 11 = 6$ $7x = -5$ $x = -\frac{5}{7}$					
$5x$	$+2$									
$-5x$	-2									
Answer: $\emptyset$ # <u>14</u> Solve for x: $\left(\frac{3-4x}{6}\right)\left(\frac{2x-3}{8}\right) \geq (2-x)$ $4(3-4x) - 3(2x-3) \geq 48-24x$ $12 - 16x - 6x + 9 \geq 48 - 24x$ $21 - 22x \geq 48 - 24x$ $21 + 24x \geq 48 - 24x$ $21 + 24x \geq 48$ $24x \geq 27$ $x \geq \frac{27}{24} = \frac{9}{8} = 1.125$	Answer: -6 # <u>24</u> Find c so that $x^2 - 10x + c$ is a perfect square trinomial. (Find c to complete the square.) Then, factor the trinomial. <table border="1" style="margin-left: auto; margin-right: auto;"><tr><td></td><td>x</td><td>-5</td></tr><tr><td>x</td><td>x^2</td><td>$-5x$</td></tr><tr><td>-5</td><td>$-5x$</td><td>25</td></tr></table> $c = 25$ $(x-5)^2$		x	-5	x	x^2	$-5x$	-5	$-5x$	25
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-5	$-5x$	25								

Answer: $x=a$ # <u>17</u> Write the equation of the line perpendicular to $2x - 3y = 12$ which through the point $(16, -7)$. $2x - 3y = 12$ $-3y = -2x + 12$ $y = \frac{2}{3}x - 4$ $y - (-7) = -\frac{3}{2}(x - 16)$ $y = -\frac{3}{2}x + 17$	Answer: $\frac{5x^6y}{3}$ # <u>2</u> Simplify $\frac{(x^{-2}y^{-1})^{-3}}{(y^6x^{-1})^{-2}}$ $\frac{x^6y^3}{y^{12}x^2}$ (x^4y^{15})
Answer: $(-\frac{1}{3}, -\frac{3}{4})$ # <u>6</u> Find the midpoint between -3π and π. $-3\pi + \pi = \frac{\quad}{2} = -3.14$	Answer: $(-\infty, 6)$ # <u>9</u> Graph $-1 \leq x \leq 3$ on a number line. Put the answer in interval notation. $[-1, 3]$
Answer: $5x+2$ # <u>21</u> Factor (bottoms up!) the trinomial $10x^2 - 13x - 3$ and find one of its factors. $\begin{array}{ c c c } \hline 5x & 1 & \\ \hline 2x & 10x & 2x \\ \hline -3 & -15x & -3 \\ \hline \end{array}$	Answer: $[13.5, \infty)$ # <u>15</u> Write the equation of the line which passes through the points $(4, 2)$ and $(-3, 1)$. $x_1y_1 \quad y_2y_2$ $m = \frac{1-2}{-3-4} \quad m = \frac{-1}{-7} = \frac{1}{7}$ $y - 2 = \frac{1}{7}(x - 4)$ $y = \frac{1}{7}x + \frac{10}{7}$ $x - 7y = -10$

Answer: x^4y^{15} # <u>3</u> Find the distance between -5 and 17. -5 → 17 (15 22)	Answer: $2x - 6$ # <u>23</u> Solve for x by first setting up two equations. Then, find the PRODUCT of your solutions. $2x + 5 = 7$ $2x + 5 = 7$ $2x = 2$ $x = 1$ $2x + 5 = -7$ $2x = -12$ $x = -6$ (-6) (1)
Answer: $-\frac{5}{7}$ # <u>13</u> Solve for x: $-2(x - 4) - \frac{1}{2}(3 - 4x) = -12$ $-2x + 8 - 1.5 + 2 = -12$ $-2x + 6.5 = -12$ $-2x = -18.5$ $x = 9.25$ no answer	Answer: $y = -1.5x + 16$ # <u>19</u> Factor (GCF) the binomial $25x^2 - 15x$ and find one of its factors. $5x$ $5x$ -3 $5x$ $25x$ -15 (5x-3)
Answer: $[-1, 3]$ # <u>10</u> The equation $(x - h)^2 + (y - k)^2 = r^2$ defines a circle with radius r and center (h, k). Find the center and radius AND graph the circle on the x-y coordinate plane. $(x - 1)^2 + (y + 3)^2 = 25$. Find the center to move on to the next question. (1, -3)	Answer: $-\pi$ # <u>7</u> If AB is a line segment with midpoint (2, 6) and endpoint A(5, 9), find the other endpoint, B. (-1, 3)

Answer: (-1, 3) # <u>8</u> Write $6 > x$ in interval notation. $(-\infty, 6)$	Answer: $7y - x = 10$ # <u>16</u> Write the equation of the line which passes through the points $(a, 2b)$ and $(a, 3b)$. $x_1 y_1 \quad x_2 y_2$ $\frac{3b - 2b}{a - a} = \frac{b}{0}$ $x = a$
Answer: $2\sqrt{13}$ # <u>5</u> Find the midpoint between $(-\frac{7}{3}, \frac{3}{4})$ and $(\frac{5}{3}, -\frac{9}{4})$. $\frac{5}{3} + (-\frac{7}{3}) = -\frac{2}{3} / 2 = -\frac{1}{3}$ $(-\frac{9}{4}) + \frac{3}{4} = -\frac{6}{4} = -\frac{3}{2}$	Answer: $5x + 1$ # <u>22</u> What is the SUM of the factors of the trinomial $x^2 - 6x + 5$? $x \begin{array}{ c } \hline x^2 \\ \hline \end{array} \begin{array}{ c } \hline x-1 \\ \hline \end{array}$ $-5 \begin{array}{ c } \hline 5 \\ \hline \end{array}$ $(x-5)(x-1)$ $2x-6$ <u>-6</u>
Answer: (1, -3) # <u>11</u> Solve for x: $\frac{2}{3}x - \frac{1}{4} = 1$ $\frac{2}{3}x = \frac{5}{4}$ $x = \frac{5}{4} \div \frac{2}{3}$ $x = \frac{15}{8}$	Answer: 22 # <u>4</u> Find the length of the line segment AB if the coordinates of the endpoints are A(-1, 5) and B(3, 11). $(-1, 5)$ $(3, 11)$ $4^2 + 6^2 = \sqrt{52} = 7.2 = 2\sqrt{13}$

Work problem #1 in the space provided. Make sure to communicate mathematically how you are arriving at your answer even if you do a lot of the work in your head or on your calculator!! Find the answer to problem #1. Number the problem in the same box #2. Work the second problem. Continue working until all are completed.

Answer: $(x-5)^2$ #1 Simplify $\frac{(5x^3)^2 y^3}{15y^2}$.

$$\frac{25x^6 y^3}{15 y^2}$$

$$\frac{5x^6 y}{3}$$

Answer: $y = -1.5x + 17$ # ____ Write the equation of the line if $m = -\frac{3}{2}$ and $b = 16$.

$$y = -\frac{3}{2}x + 16$$

Answer: $5x - 3$ # ____ Factor (difference of squares) the binomial $25x^2 - 4$ and find one of its factors.

$$(5x-2)(5x+2)$$

Answer: $\frac{15}{8}$ # ____ Solve for x : $\frac{x-1}{3} + \frac{x+5}{4} = \frac{1}{2}$

$$\frac{4x-4}{12} + \frac{3x+15}{12} = \frac{6}{12}$$

$$7x + 11 = 6$$

$$7x = -5$$

$$x = -\frac{5}{7}$$

Answer: $\emptyset$ # 12 Solve for x : $\frac{3-4x}{6} - \frac{2x-3}{8} \geq 2-x$

$$\frac{12-16x}{24} - \frac{6x-9}{24} \geq 2-x$$

$$-22x+21 \geq 48-24x$$

$$-27 \geq -24x$$

$$13.5 \leq x$$

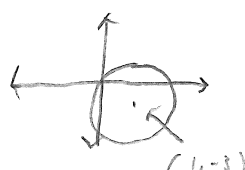
Answer: -6

____ Find c so that $x^2 - 10x + c$ is a perfect square trinomial. (Find c to complete the square.) Then, factor the trinomial.

$$\begin{array}{r} x-5 \\ x \quad | \quad x-5x \\ -5 \quad | \quad 5x \quad c \end{array}$$

$$25$$

Answer: $x=a$ # ____ Write the equation of the line perpendicular to $2x - 3y = 12$ which through the point $(16, -7)$. $2x - 3y = 12$ $y + 7 = -\frac{3}{2}(x - 16)$ $-3y = 12 - 2x$ $y + 7 = -\frac{3}{2}x + 24$ $y = -\frac{3}{2}x - 4$ $y = -\frac{3}{2}x + 17$	Answer: $\frac{5x^6y}{3}$ # ____ Simplify $\frac{(x^{-2}y^{-1})^{-3}}{(y^6x^{-1})^{-2}}$ $\frac{x^6 y^3}{y^{-12} x^2}$ $x^4 y^{15}$ $\frac{x^6 y^3 y^{12}}{x^2}$
Answer: $(-\frac{1}{3}, -\frac{3}{4})$ # ____ Find the midpoint between -3π and π. $-3\pi + \pi = -2\pi$ $-\pi$	Answer: $(-\infty, 6)$ # ____ Graph $-1 \leq x \leq 3$ on a number line. Put the answer in interval notation. $[-1, 3]$ 
Answer: $5x+2$ # ____ Factor (bottoms up!) the trinomial $10x^2 - 13x - 3$ and find one of its factors. $(5x+2)(2x-3)$	Answer: $[13.5, \infty)$ # ____ Write the equation of the line which passes through the points $(4, 2)$ and $(-3, 1)$. $m = \frac{1 - 2}{-3 - 4} = \frac{-1}{-7} = \frac{1}{7}$ $2 = \frac{1}{7}x + b$ $y = \frac{1}{7}x + \frac{13}{7}$

Answer: x^4y^{15} # ____ Find the distance between -5 and 17. 22	Answer: $2x - 6$ # ____ Solve for x by first setting up two equations. Then, find the PRODUCT of your solutions. $2x + 5 = 7$ $2x + 5 = 7$ $2x + 5 = -7$ $2x = 2$ $2x = -12$ $x = 1$ $x = -6$ -7
Answer: $-\frac{5}{7}$ # ____ Solve for x: $-2(x - 4) - \frac{1}{2}(3 - 4x) = -12$ $-2x + 4 - 1.5 + 2x = -12$ $0x + 2.5 = -12$	Answer: $y = -1.5x + 16$ # ____ Factor (GCF) the binomial $25x^2 - 15x$ and find one of its factors. $5x - 3$
Answer: $[-1, 3]$ # ____ The equation $(x - h)^2 + (y - k)^2 = r^2$ defines a circle with radius r and center (h, k). Find the center and radius AND graph the circle on the x-y coordinate plane. $(x - 1)^2 + (y + 3)^2 = 25$. Find the center to move on to the next question. $(1, -3)$ - center $r = 5$ 	Answer: $-\pi$ # ____ If AB is a line segment with midpoint (2, 6) and endpoint A(5, 9), find the other endpoint, B. $(-1, 12)$

Answer: $(-1, 3)$ # _____ Write $6 > x$ in interval notation. $(-\infty, 6)$	Answer: $7y - x = 10$ # _____ Write the equation of the line which passes through the points $(a, 2b)$ and $(a, 3b)$. $x = a$
Answer: $2\sqrt{13}$ # _____ Find the midpoint between $(-\frac{7}{3}, \frac{3}{4})$ and $(\frac{5}{3}, -\frac{9}{4})$. $(-\frac{1}{3}, -\frac{2}{5})$	Answer: $5x+1$ # _____ What is the SUM of the factors of the trinomial $x^2 - 6x + 5$? $(x-1)(x-5)$ -6
Answer: $(1, -3)$ # _____ Solve for x: $\frac{2}{3}x - \frac{1}{4} = 1$ 5	Answer: 22 # _____ Find the length of the line segment AB if the coordinates of the endpoints are $A(-1, 5)$ and $B(3, 11)$. $2\sqrt{13}$

Work problem #1 in the space provided. Make sure to communicate mathematically how you are arriving at your answer even if you do a lot of the work in your head or on your calculator!! Find the answer to problem #1. Number the problem in the same box #2. Work the second problem. Continue working until all are completed.

Answer: $(x-5)^2$ #1 Simplify $\frac{(5x^3)^2 y^3}{15y^2}$.

$$\frac{5x^3 \cdot 5x^3 \cdot y^2}{15y} \rightarrow \frac{25x^6 \cdot y^2}{15y} \rightarrow \frac{5x^6 y^2}{3y}$$

$$\frac{5x^6 y}{3}$$

Answer: $y = -1.5x + 17$ # ____ Write the equation of the line if $m = -\frac{3}{2}$ and $b = 16$.

$$y = -\frac{3}{2}x + 16$$

Answer: $5x - 3$ # ____ Factor (difference of squares) the binomial $25x^2 - 4$ and find one of its factors.

$$\begin{array}{r|rr} 5x & 2 \\ \hline 25x^2 & 10x \\ \hline -2 & -4 \end{array}$$

$$(5x-2)(5x+2)$$

Answer: $\frac{15}{8}$ # ____ Solve for x : $\frac{x-1}{3} + \frac{x+5}{4} = \frac{1}{2}$

$$\frac{x-1}{3} + \frac{x+5}{4} = \frac{1}{2}$$

$$\frac{4x-4}{12} + \frac{3x+15}{12} = \frac{1}{2}$$

$$\frac{7x+11}{12} = \frac{1}{2}$$

$$7x+11 = 6$$

$$7x = -5$$

$$x = -\frac{5}{7}$$

Answer: $\emptyset$ # ____ Solve for x : $\frac{3-4x}{6} - \frac{2x-3}{8} \geq 2-x$

$$\frac{3-4x}{6} - \frac{2x-3}{8} \geq 2-x$$

$$\frac{+12-16x}{24} + \frac{-6x+9}{24} \geq 2-x$$

$$\frac{-22x+21}{24} \geq 2-x$$

$$-22x+21 \geq 48-24x$$

$$+22x$$

$$21 \geq 48-2x$$

$$-12x$$

$$21 \geq 48-2x$$

$$-27 \geq -2x$$

$$x \geq \frac{27}{2}$$

Answer: -6 # ____ Find c so that $x^2 - 10x + c$ is a perfect square trinomial. (Find c to complete the square.) Then, factor the trinomial.

$$\begin{array}{r|rr} x & -5 \\ \hline x^2 & -10x \\ \hline -5 & +c \end{array}$$

$$25$$

Answer: $x=a$

_____ Write the equation of the line perpendicular to $2x - 3y = 12$ which through the point $(16, -7)$.

$$2x - 3y = 12$$

$$2x = 12 + 3y$$

$$x = \frac{12 + 3y}{2}$$

$$x = 6 + \frac{3}{2}y$$

$$-\frac{3}{2}(-7) + b = 16$$

$$\frac{21}{2} + b = 16$$

$$b = 16 - \frac{21}{2}$$

$$b = \frac{32}{2} - \frac{21}{2}$$

$$b = \frac{11}{2}$$

$$y - \frac{3}{2}x = -\frac{11}{2}$$

Answer: $\frac{5x^6y}{3}$

_____ Simplify $\frac{(x^{-2}y^{-1})^{-3}}{(y^6x^{-1})^{-2}}$

$$\frac{(x^{-2}y^{-1})^{-3}}{(y^6x^{-1})^{-2}}$$

$$\frac{x^6y^3}{y^{-12}x^2}$$

$$\frac{x^4y^3}{x^2y^{-12}}$$

$$\frac{x^4y^3}{x^2y^{-12}} = x^2y^{15}$$

Answer: $(-\frac{1}{3}, -\frac{3}{4})$

_____ Find the midpoint between -3π and π .

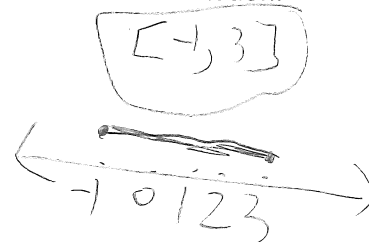
$$\frac{-3\pi + \pi}{2}$$

$$\frac{-2\pi}{2}$$

$$-\pi$$

Answer: $(-\infty, 6)$

_____ Graph $-1 \leq x \leq 3$ on a number line. Put the answer in interval notation.



Answer: $5x+2$

_____ Factor (bottoms up!) the trinomial $10x^2 - 13x - 3$ and find one of its factors.

$$10x^2 - 13x - 3$$

$$(2x-3)(5x+1)$$

Answer: $[13.5, \infty)$

_____ Write the equation of the line which passes through the points $(4, 2)$ and $(-3, 1)$.

$$y - 2 = \frac{1}{7}(x - 4)$$

$$y - 2 = \frac{1}{7}x - \frac{4}{7}$$

$$y = \frac{1}{7}x - \frac{4}{7} + 2$$

$$y = \frac{1}{7}x - \frac{4}{7} + \frac{14}{7}$$

$$y = \frac{1}{7}x + \frac{10}{7}$$

Answer: $x^4 y^{15}$ # ____ Find the distance between -5 and 17. <i>Handwritten: 22</i>	Answer: $2x - 6$ # ____ Solve for x by first setting up two equations. Then, find the PRODUCT of your solutions. $2x + 5 = 7$ <i>Handwritten work:</i> $2x + 5 = 7$ $2x = 2$ $x = 1$ $-(2x + 5) = 7$ $-2x - 5 = 7$ $-2x = 12$ $x = -6$ Product: $1 \cdot (-6) = -6$
Answer: $-\frac{5}{7}$ # ____ Solve for x: $-2(x - 4) - \frac{1}{2}(3 - 4x) = -12$ <i>Handwritten work:</i> $-2x + 8 - \frac{3}{2} + 2x = -12$ $-\frac{3}{2} = -20$ $-3 = -40$ $-3 + 40 = -40 + 40$ $37 = 0$ DNE	Answer: $y = -1.5x + 16$ # ____ Factor (GCF) the binomial $25x^2 - 15x$ and find one of its factors. <i>Handwritten work:</i> $5x^2$ $5x(5x - 3)$ $5x$ $25x(x - 1)$
Answer: $[-1, 3]$ # ____ The equation $(x - h)^2 + (y - k)^2 = r^2$ defines a circle with radius r and center (h, k). Find the center and radius AND graph the circle on the x-y coordinate plane. $(x - 1)^2 + (y + 3)^2 = 25$. Find the center to move on to the next question. <i>Handwritten work:</i> $5 = r$ $C = (1, -3)$	Answer: $-\pi$ # ____ If AB is a line segment with midpoint (2, 6) and endpoint A(5, 9), find the other endpoint, B. <i>Handwritten work:</i> (a, b) $\frac{5+a}{2} = 2$ $5+a = 4$ $a = -1$ $\frac{9+b}{2} = 6$ $9+b = 12$ $b = 3$ B = (-1, 3) Length AB: $\sqrt{(-1-5)^2 + (3-9)^2} = \sqrt{36+36} = \sqrt{72} = 6\sqrt{2}$ Area of circle: $\pi r^2 = \pi (6\sqrt{2})^2 = 72\pi$

Answer: $(-1, 3)$

✓ Write $6 > x$ in interval notation.

$$6 > x$$

$$(6, -\infty)$$

Answer: $7y - x = 10$

Write the equation of the line which passes through the points $(a, 2b)$ and $(a, 3b)$.

$$y = mx + b$$

$$9 = a$$

$$\frac{2b}{2} = \frac{m(a)}{2} + c = b$$

$$\frac{3b}{2} = \frac{m(a)}{2} + c$$

Answer: $2\sqrt{13}$

Find the midpoint between $(-\frac{7}{3}, \frac{3}{4})$ and $(\frac{5}{3}, -\frac{9}{4})$.

$$-\frac{7}{3} + \frac{5}{3} \quad -\frac{9}{4} + \frac{3}{4}$$

$$-\frac{2}{3} \quad -\frac{1}{4} \quad -\frac{6}{4} \quad -\frac{3}{2}$$

$$(-\frac{1}{3}, -\frac{3}{4})$$

Answer: $5x+1$

What is the SUM of the factors of the trinomial $x^2 - 6x + 5$?

$$\begin{array}{r|rr} x & -1 & \\ \hline x & x^2 & -1x \\ -5 & -5x & 5 \\ \hline (x-1)(x-5) & & \\ 2x-6 & & \end{array}$$

Answer: $(1, -3)$

Solve for x: $\frac{2}{3}x - \frac{1}{4} = 1$

$$\frac{8}{12}x - \frac{1}{4} = 1$$

$$\frac{2}{3}x - \frac{1}{4} = 1$$

$$\frac{2}{3}x - \frac{1}{4} = 1$$

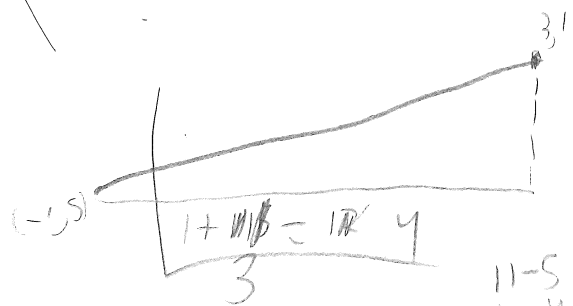
$$\frac{2}{3}x = 1 + \frac{1}{4}$$

$$\frac{2}{3}x = \frac{5}{4}$$

$$\frac{15}{6} \cdot \frac{3}{2} = \frac{5}{2}$$

Answer: 22

Find the length of the line segment AB if the coordinates of the endpoints are $A(-1, 5)$ and $B(3, 11)$.



$$2\sqrt{13}$$

$$\begin{aligned} 4^2 + 6^2 &= 16 + 36 \\ &= 52 \\ \sqrt{52} &= 2\sqrt{13} \end{aligned}$$